



Avaya Solution & Interoperability Test Lab

Avaya Dialog Designer – TelephonyWebService Sample Application Documentation – Issue 1.2

Abstract

This document describes the call flow, functionality and implementation of the Telephony Web Service ('TelephonyWebService') sample application designed using Avaya Dialog Designer (DD). The application describes how to invoke a Telephony Web Service Description Language (WSDL) file, hosted by Avaya Application Enablement Server (AES) 4.0. Using this Web Service, the application performs Make Call, Single Step Conference Call, Single Step Transfer Call and Disconnect Active Call operations. The application is menu driven where the user needs to select the operation to perform. The outcome of the operation is announced to the user. The application also demonstrates the exception handling mechanism supported by DD 4.0.

The TelephonyWebService application has been validated to work correctly with Dialog Designer 4.1.7, Voice Portal 4.1 and Interactive Response 3.0, as well as Avaya Aura® Orchestration Designer 6.0 and Avaya Aura® Experience Portal 6.0

Table of Contents

1. Introduction and Scope	3
1.1. Equipment and Software Used:	4
2. <i>TelephonyWebService</i> Application	5
2.1. Call Flow Overview	5
2.2. List of Avaya Application Enablement Services (AES) Web Services Used	7
3. Building the <i>TelephonyWebService</i> Dialog Designer application using Avaya Application Enablement Services (AES) Web Services	8
3.1. Creating Phantom extensions on the Avaya Communication Manager	13
4. Creating a .WAR (Deployment) file for the <i>TelephonyWebServices</i> Dialog Designer Application	14
5. Deploying the <i>TelephonyWebService</i> Dialog Designer application on a local Tomcat Web Server	19
6. Deploying the <i>TelephonyWebService</i> application	22
6.1. Avaya Voice Portal Platform	22
6.2. Avaya Interactive Response	25
6.3. Testing the <i>TelephonyWebService</i> application	28
7. References	32

Table of Figures

Figure 1: Network configuration for testing Avaya Dialog Designer applications	3
--	---

Table of Tables

Table 1: Equipment and Software Version Information	4
Table 2: Telephony Web Service Requests	7

1. Introduction and Scope

The 'TelephonyWebService' application demonstrates the following:

1. Access the Avaya Application Enablement Services (AES) Telephony Web Service (WSDL) file using the Dialog Designer Web Services connector.
2. Perform basic 3rd party call control features like Make Call, Single Step Conference Call, Single Step Transfer Call, and Disconnect Active Call on Communication Manager.

The configuration for deploying and testing an Avaya Dialog Designer Telephony Web Services application using Avaya Voice Portal is as shown below.

Note: The setup and application call flow will remain the same if an Avaya Interactive Response system is used instead of the Avaya Voice Portal System shown below.

The TelephonyWebService application has been validated to work correctly with Dialog Designer 4.1.7, Voice Portal 4.1 and Interactive Response 3.0, as well as Avaya Aura® Orchestration Designer 6.0 and Avaya Aura® Experience Portal 6.0

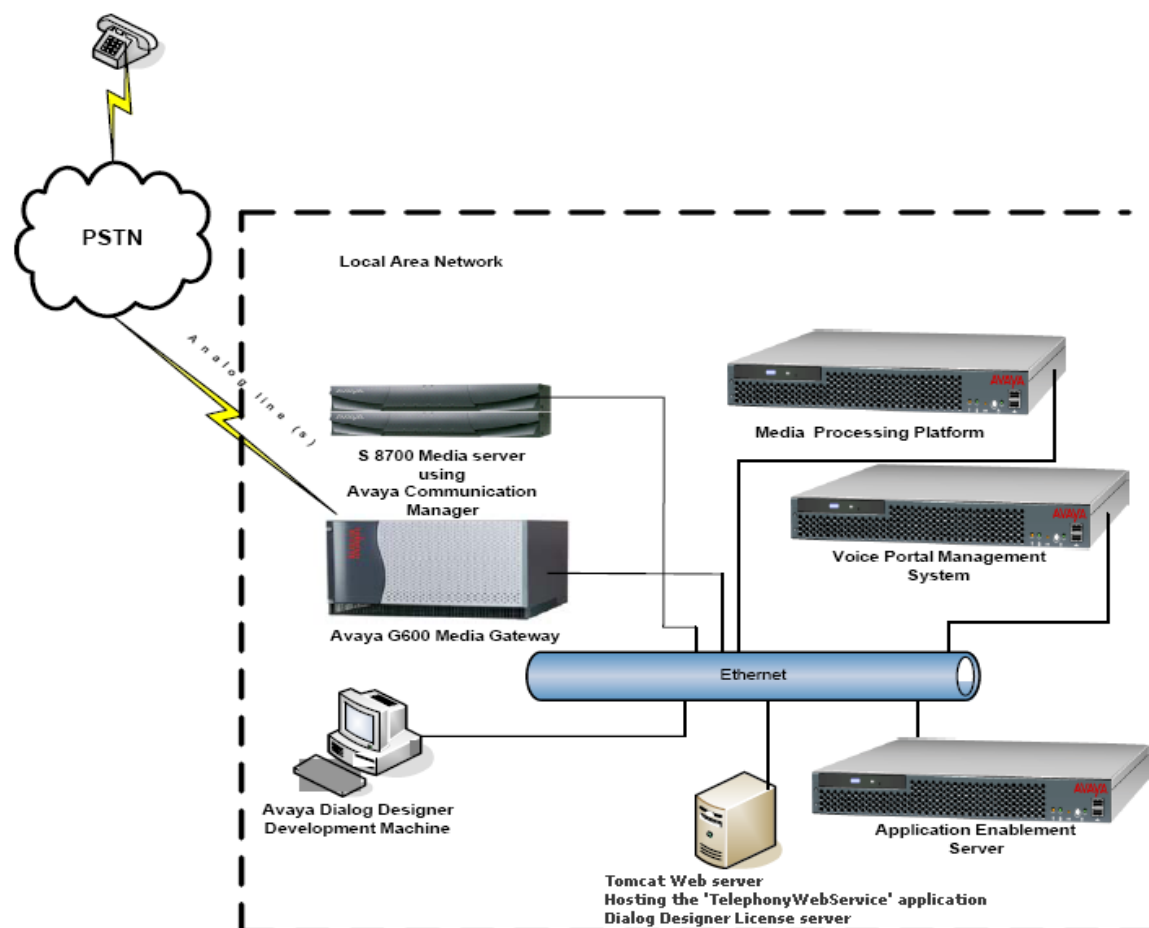


Figure 1: Network configuration for testing Avaya Dialog Designer applications

1.1. Equipment and Software Used:

Table 1 shows the equipment and software version information

Equipment	Software
Avaya Voice Portal Management System (VPMS)	V 4.0.0.0.2901
Avaya Media Processing Platform (MPP) on a S8500B Media Server	V 4.0.0.0.2901
Avaya Communication Manager Avaya S8300 Media Server Avaya G700 Media Gateway	R014x.00.0.730.2
Avaya Dialog Designer	4.0.7
Apache Tomcat Web Server	5.5.23
Application Enablement Server (AES) on a S8500C Media Server	4.0
Desktop PC Pentium 4 1 GB RAM, 1.70GhZ 40 GB Hard Disk 100 MHz network interface card	Windows XP Professional Version 2002 – Service Pack 2 Avaya WebLM 4.0

Table 1: Equipment and Software Version Information

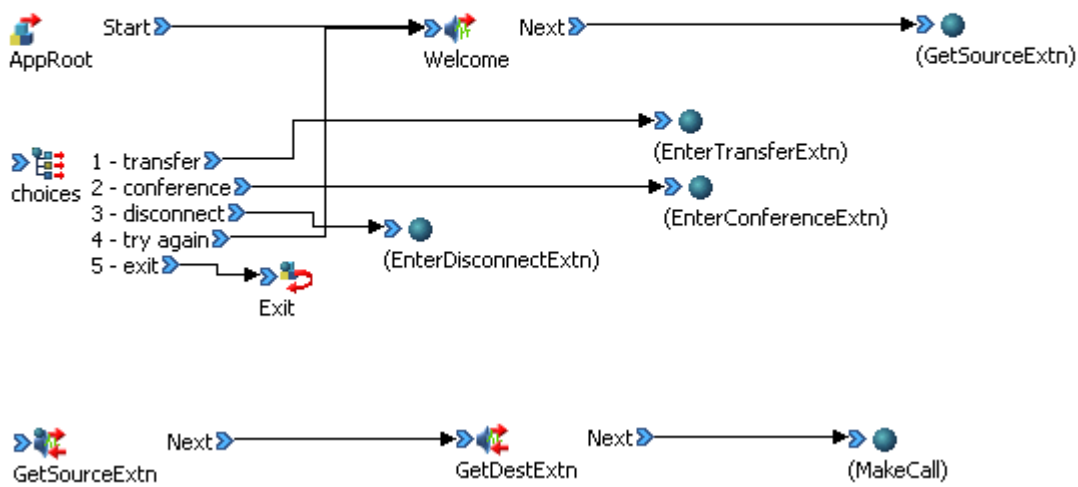
2. TelephonyWebService Application

2.1. Call Flow Overview

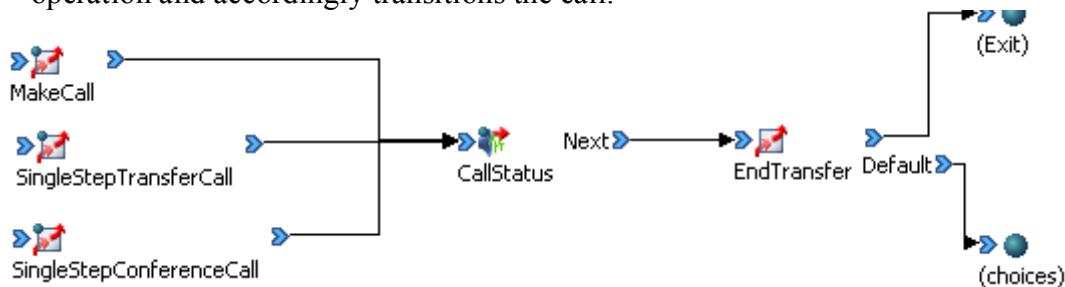
The call flow for the *TelephonyWebService* application can be broken down into following steps:

1. The application answers an incoming call and prompts the caller to provide two extensions for **makeCall** operation. Once caller enters the extensions, the application invokes the makeCall operation available in the Telephony Web Service. Depending upon the success or failure of makeCall, the status is announced to the caller.

The Telephony Web Service Application prompts the caller to first initiate a call. Once the call is made successful, the caller is provided the option to choose one of single step transfer, single step conference, disconnect, make call operation or exit the application.



2. After the **makeCall** operation, the caller is prompted to choose one of the following five options: Single Step Transfer Call, Single Step Conference Call, Disconnect Call, Try Again and Exit as shown above. The callStatus data node evaluates the result of the operation and accordingly transitions the call.



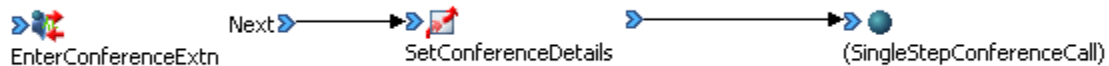
3. Single Step Transfer Call option:

On selecting this option, the caller is prompted to enter an extension to which the call has to transfer. Then the Telephony web service is invoked with the transfer method selected. If the transfer is successful, the current existing call gets terminated and the call gets transferred to the extension specified. The success or failure of the transfer operation is announced to the caller and the application exits.



4. Single Step Conference Call option: If

The caller selects single step conference call option, the caller is prompted to enter an extension to add in the conference. The telephony web service is invoked to conference in the extension specified. The success or failure of operation is announced to the caller. If the operation is successful the extension specified gets added to the conference.



5. Disconnect Call

The disconnect call option prompts the caller to enter an extension. If the extension is involved in an active call, the call gets disconnected on invoking the web service.



2.2. List of Avaya Application Enablement Services (AES) Web Services Used

Table 2 below displays the list functionality supported by AES Telephony web service.

Service	Description
Make Call	Originates an outbound call from one party to another
Single Step Conference Call	Conferences in a new party to the existing active call
Single Step Transfer Call	Transfers the remote party on the current active call to the specified destination
Disconnect Active Call	Disconnects the specified extension from the active call
Answer Alerting Call	Answers an alerting call at the specified destination
Attach	Allows an application to authenticate the user and explicitly establish a session before invoking a telephony request.
Release	Explicitly ends the session

Table 2: Telephony Web Service Requests

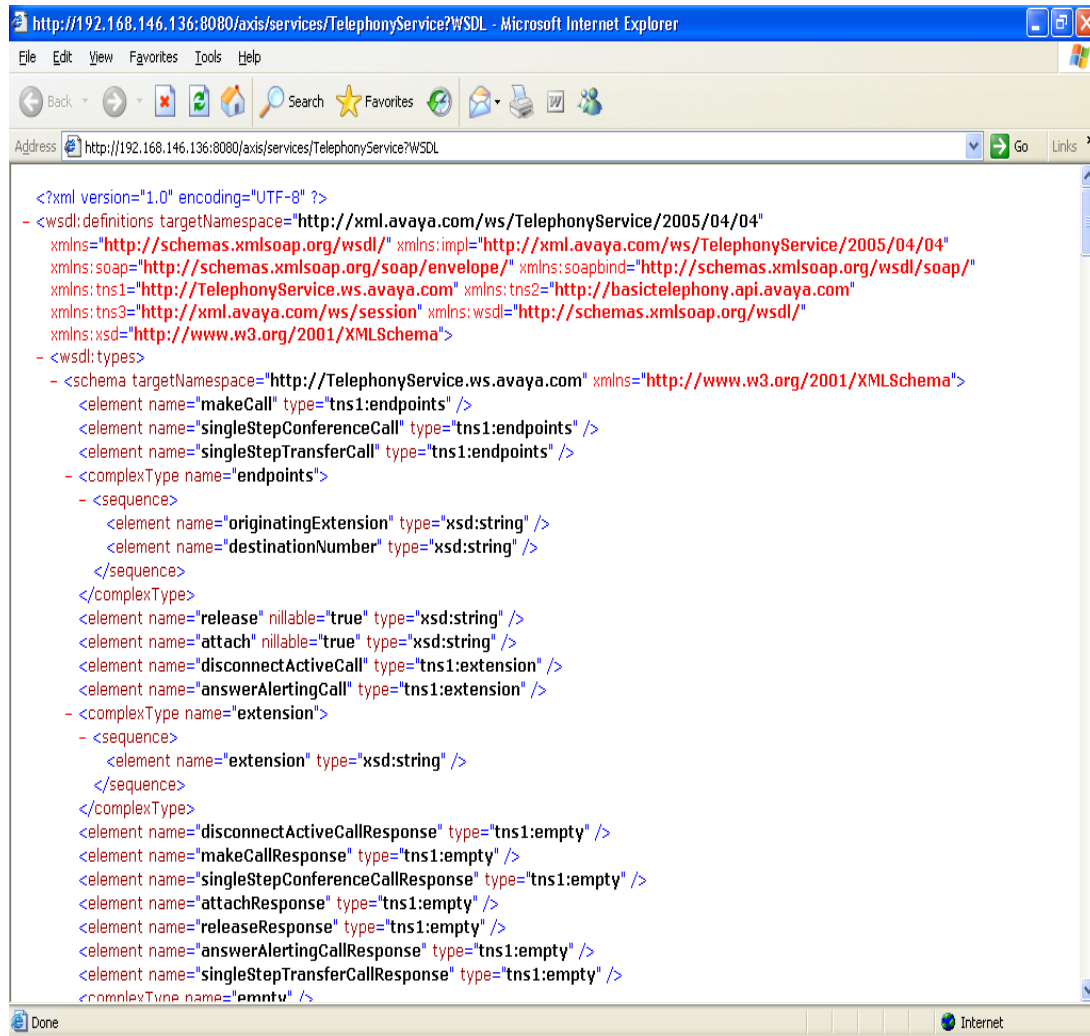
Note:

1. View a list of the all the AES web services by entering the URL
<http://<AES Server Name>:8080/axis/services>
2. The Telephony web service (describing the above operations) can be accessed at
<http://<AES Server Name>:8080/axis/services/TelephonyService?wsdl>

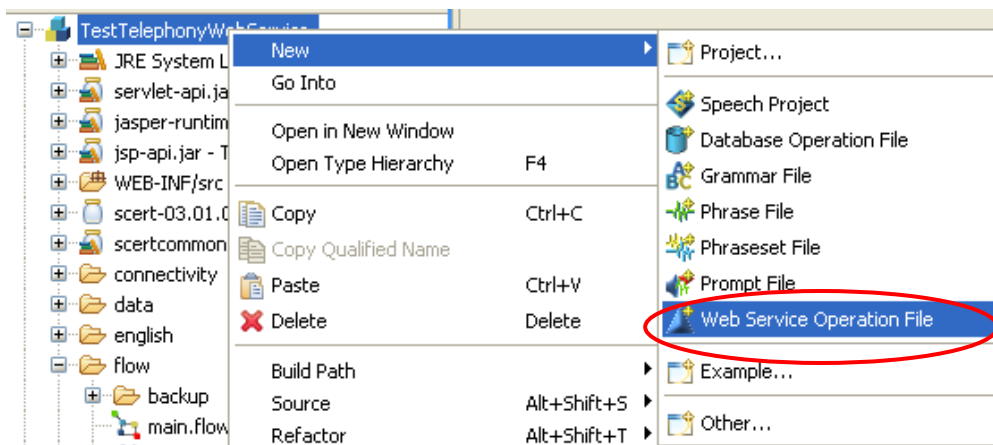
Refer to [1] for additional information on AES web services.

3. Building the *TelephonyWebService* Dialog Designer application using Avaya Application Enablement Services (AES) Web Services

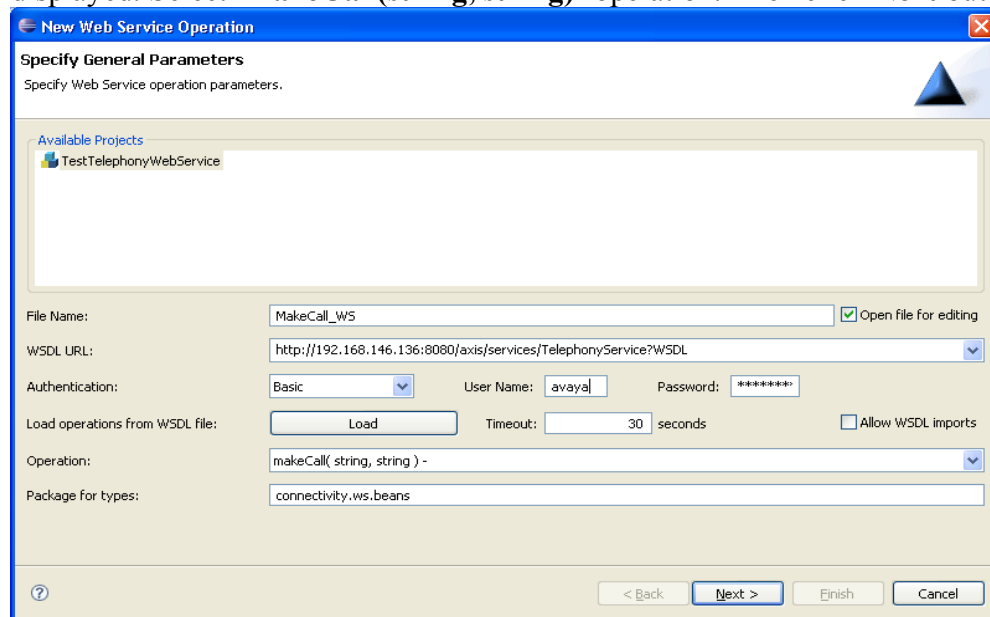
Step 1. Open a Web browser and access the following URL:
http://<AESServerName>:8080/axis/services/TelephonyService?wsdl where **ServerName** is the AES server name or IP Address. The WSDL is displayed as shown below.



Step 2. Create a Speech project. Right click on the project and select **New**. Choose **Web Service Operation File** as shown below.



Step 3. Enter appropriate name in the **File Name** column and enter the Telephony Web Service URL in the **WSDL URL** column as shown below. In the **Authentication** drop down select '**Basic**'. In the **User Name** and **Password** field enter the **CT user name** and **password**. Then click on **Load** Button, if the load is successful, a drop down list containing the operation supported by the web service is displayed. Select '**makeCall(string, string)**' operation. Then click **Next** button.



Step 4. Click on **Auto Create** to automatically create the variables `originatingExtension` and `destinationNumber`. Alternatively variables can also be created by dragging a **simple variable** in *flow/project.variables* pane. Then corresponding variables can be selected from the *Variable Name* drop down list. Click on **Next** to proceed.

Parameter Name	Parameter Type	Variable Name	Variable Field	Auto Create	Use Java Obj
originatingExtension	string			☒	☐
destinationNumber	string			☒	☐

Step 5. The next screen is for mapping output parameters. Since this web service does not return anything, click on **Finish**.

Parameter Name	Parameter Type	Variable Name	Variable Field	Auto Create	Use Java Obj
----------------	----------------	---------------	----------------	-------------	--------------

Step 6. To maintain a session, the sessionID parameter needs to be added in the Soap Headers. To add the sessionID parameter click on **Add** button, enter the name '**sessionID**' in the Name field. Create a simple variable sessionidentifier using **Flow -> project variables** palette. Select the created variable from the variable drop down list. For getting new sessionID, select '**In**' from the drop down list of Direction column. And to reuse the existing sessionID, select '**Out**' from the same drop down list. Save the WSOP file.

Headers
Both SOAP and HTTP headers can be send and received during the web service operation

HTTP Headers

Name	Variable	Field	Direction

SOAP Headers

Name	Variable	Field	Direction	Must Understa
sessionID	sessionidentifier		In	

WSOP Editor

Step 7. Select application flow view and drag the **Data node** from the Application Items section in the palette as shown below. Rename the node.

Palette

- Select
- Connection
- Label
- Templates
- Announce
- Prompt & Collect
- Disconnect
- Application Items
- Data
- Tracking
- Avaya
- CollectTicketInfo
- StadiumDirections

Application Flow

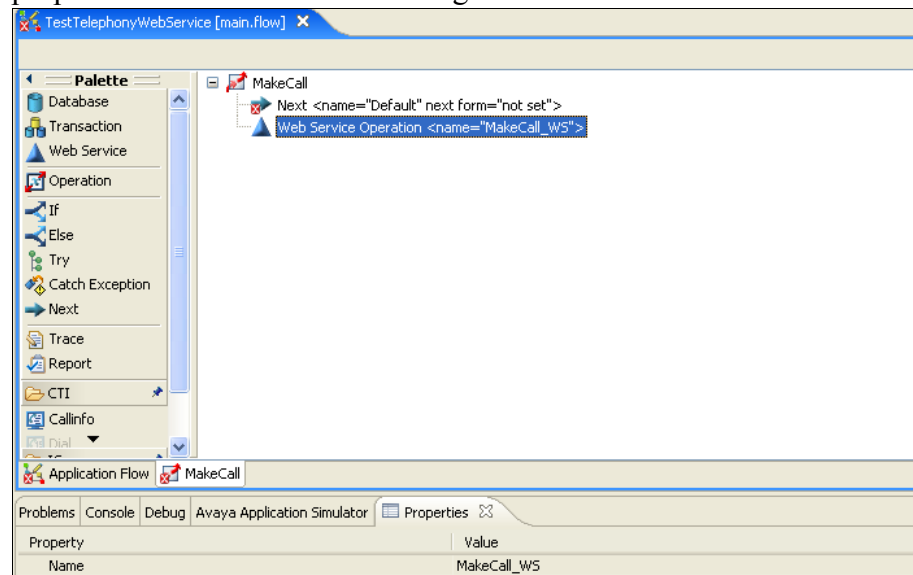
AppRoot Start

MakeCall

Properties

Property	Value
Name	MakeCall
Comments	
Location	(121, 38)

Step 8. Double click on the *Data* node to open it. Drag the *Web Service* node from the palette and select the WSOP file that was created earlier from the drop down list in properties window. Save the settings.



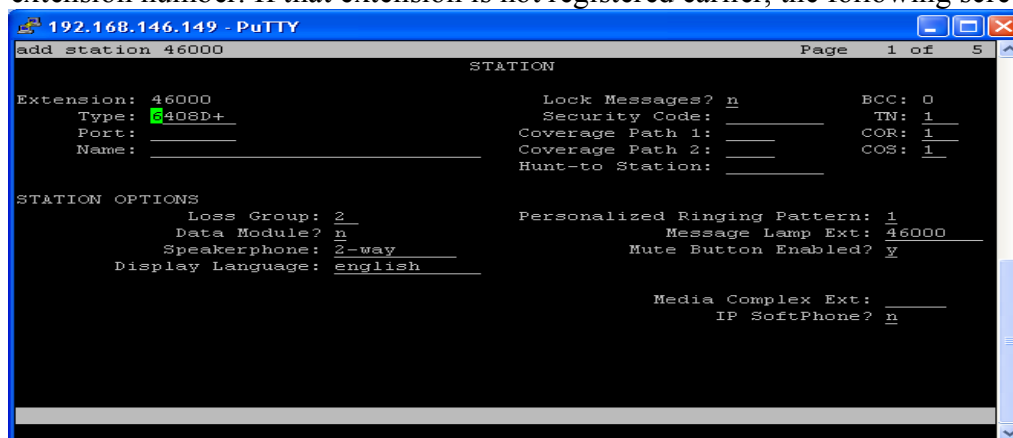
Step 9. Similarly the TransferCall, DisconnectCall, AnswerCall operations from Telephony Web Service can be used in any Speech Application. The corresponding operation needs to be selected while creating the WSOP file.

3.1. Creating Phantom extensions on the Avaya Communication Manager

This section describes how to create phantom extensions on the Avaya Communication Manager. Refer to [2] for additional information.

Note: This is an optional step. The user can enter the phantom extension created below when prompted by the system for a source extension. This allows a MakeCall operation to complete the call between the source (phantom) extension and the destination number provided by the user. In this scenario, the phantom extension acts as the caller's secret (virtual) phone extension to make calls remotely (by calling into the system).

Step 1. Log in to SAT terminal and enter command “**add station xxxx**” where **xxxx** is the extension number. If that extension is not registered earlier, the following screen gets displayed.



192.168.146.149 - PuTTY

add station 46000

Page 1 of 5

STATION

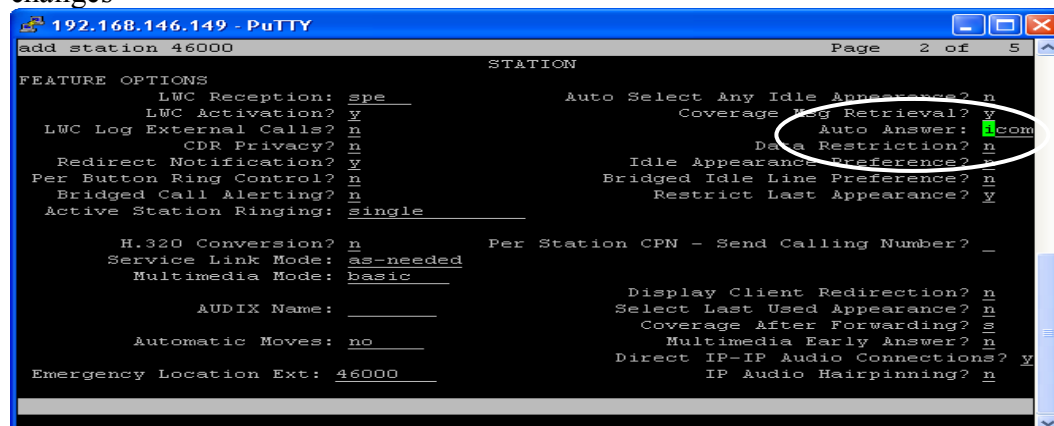
Extension: 46000 Lock Messages? n BCC: 0
Type: 640SD+ Security Code: TN: 1
Port: Coverage Path 1: COR: 1
Name: Coverage Path 2: COS: 1
 Hunt-to Station:

STATION OPTIONS

Loss Group: 2 Personalized Ringing Pattern: 1
Data Module? n Message Lamp Ext: 46000
Speakerphone: 2-way Mute Button Enabled? y
Display Language: english

Media Complex Ext:
IP SoftPhone? n

Step 2. Locate the **Auto Answer** setting in the Station form. Change the value of the Auto Answer setting to '**icom**'. The station will auto answer all subsequent incoming calls. Save the changes



192.168.146.149 - PuTTY

add station 46000

Page 2 of 5

STATION

FEATURE OPTIONS

LWC Reception: spe Auto Select Any Idle Appearance? n
LWC Activation? y Coverage Req Retrieval? y
LWC Log External Calls? n Auto Answer: icom
 Data Restriction? n
CDR Privacy? n Idle Appearance Preference? n
Redirect Notification? y Bridged Idle Line Preference? n
Per Button Ring Control? n Restrict Last Appearance? y
Bridged Call Alerting? n

Active Station Ringing: single

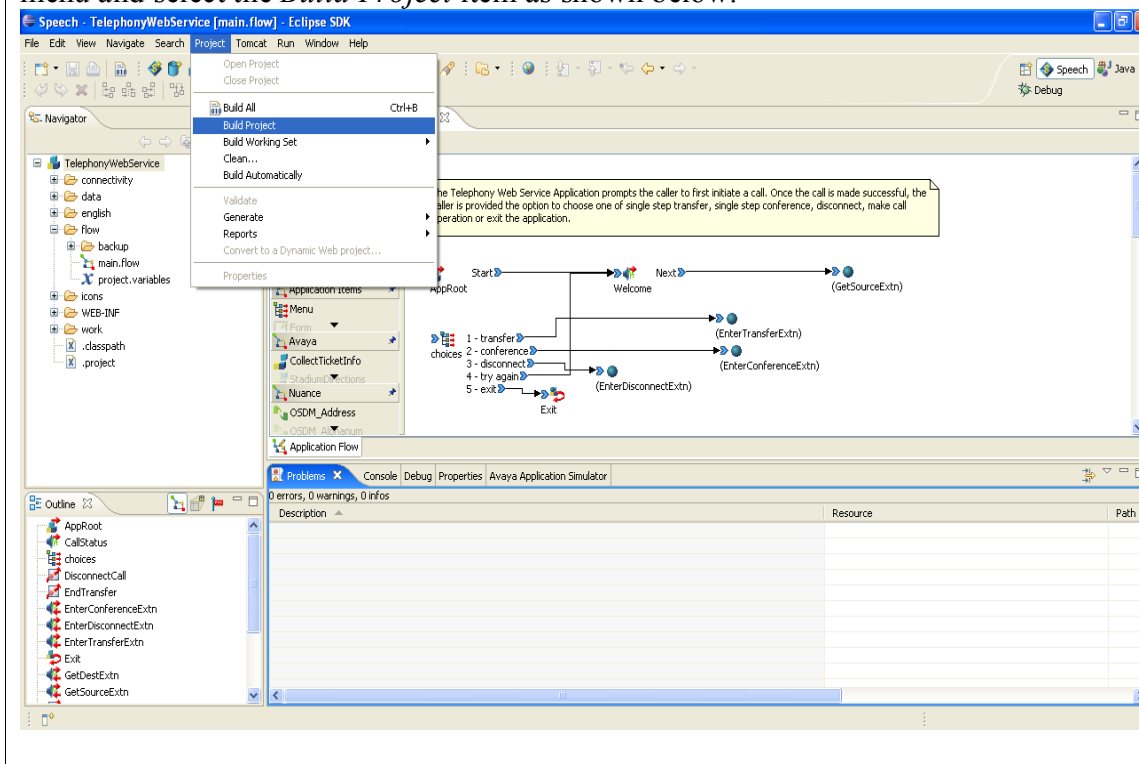
H.320 Conversion? n Per Station CPN - Send Calling Number? _
Service Link Mode: as-needed
Multimedia Mode: basic

AUDIX Name: Display Client Redirection? n
Automatic Moves: no Select Last Used Appearance? n
 Coverage After Forwarding? n
Emergency Location Ext: 46000 Multimedia Early Answer? n
 Direct IP-IP Audio Connections? y
 IP Audio Hairpinning? n

4. Creating a .WAR (Deployment) file for the TelephonyWebServices Dialog Designer Application

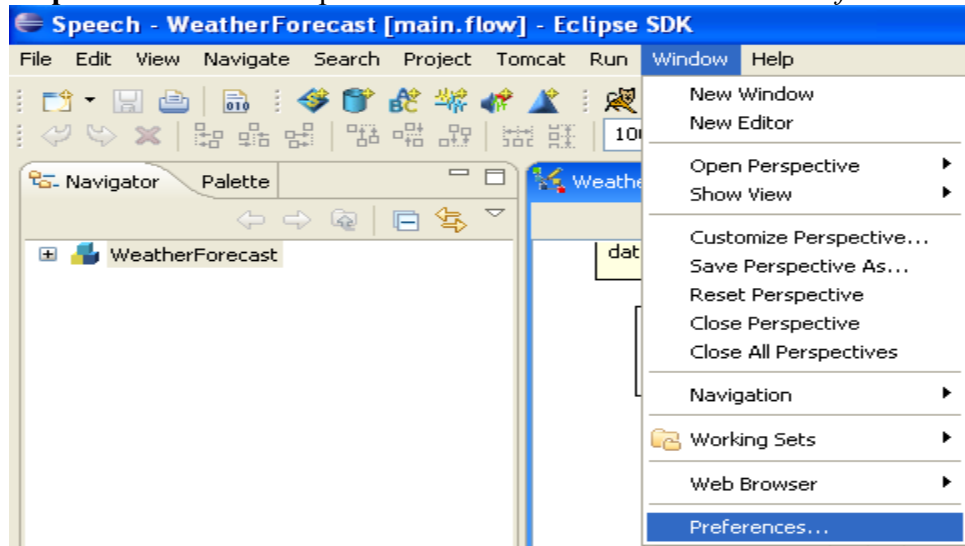
Note: It is not necessary to create a .WAR file for deployment, as each Dialog Designer application's context path is available to the local Tomcat server. In the event of deploying a .WAR file to local tomcat first delete the application from Dialog Designer, remove the context definition for that application and then follow the instructions below to deploy the application

Step 1. Highlight the project in the *Navigator* view and click on *Project* in the top -level menu and select the *Build Project* item as shown below.

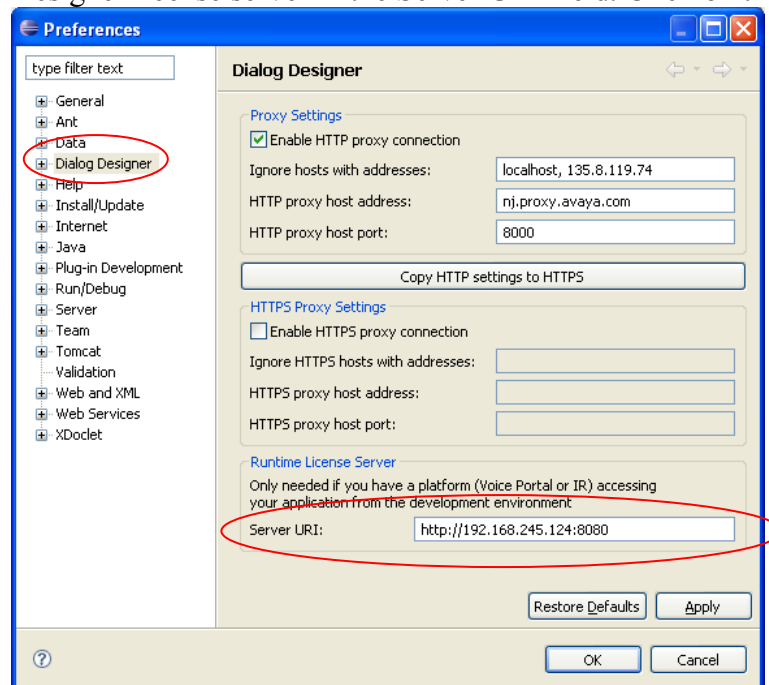


Important: Setting the Dialog Designer License server URI

Step 2. Click on the top level menu and click on *Window->Preferences* as shown below.

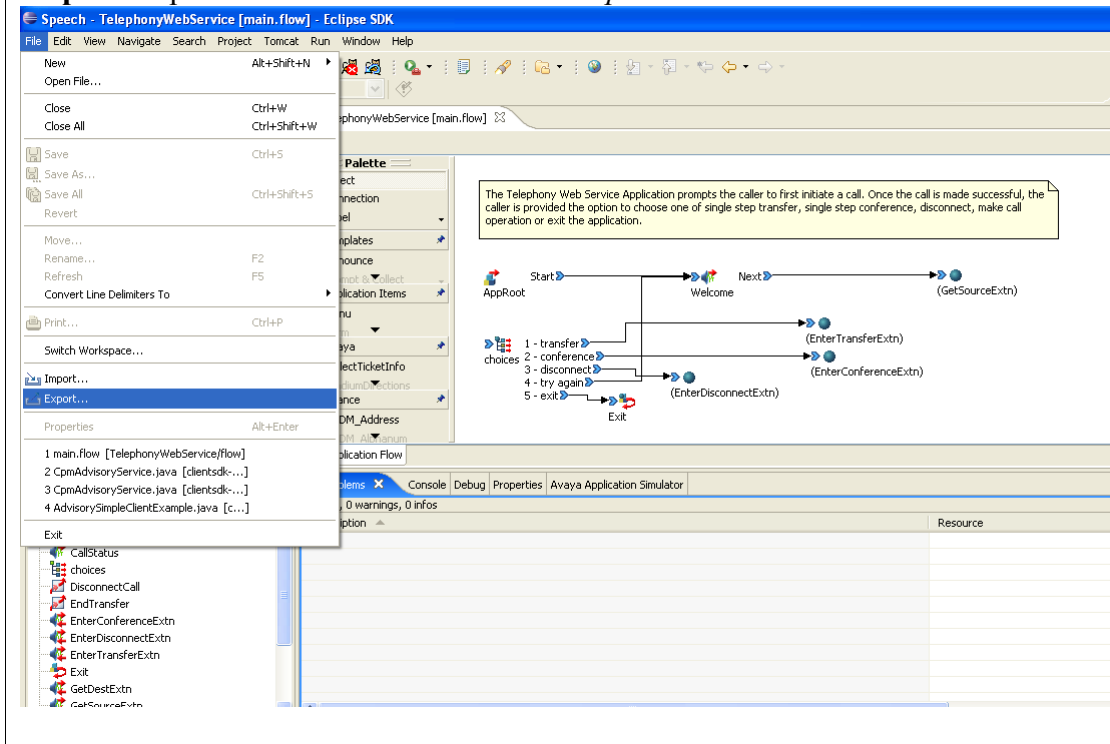


Step 3. Select Dialog Designer in the left hand pane. Enter the IP address of the Dialog Designer license server in the Server URI field. Click on the OK button.

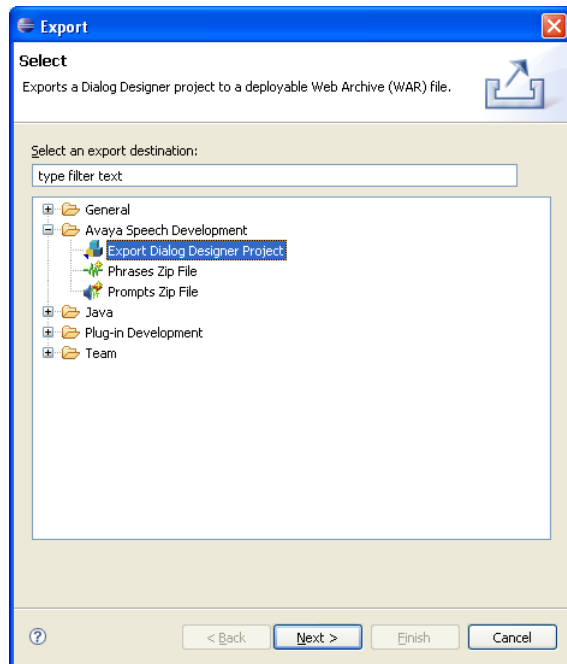


Note: If a proxy connection is needed for the application to access the internet, check the *Enable HTTP proxy connection* box and complete the remaining fields as shown above. Refer to Dialog Designer Help for additional information.

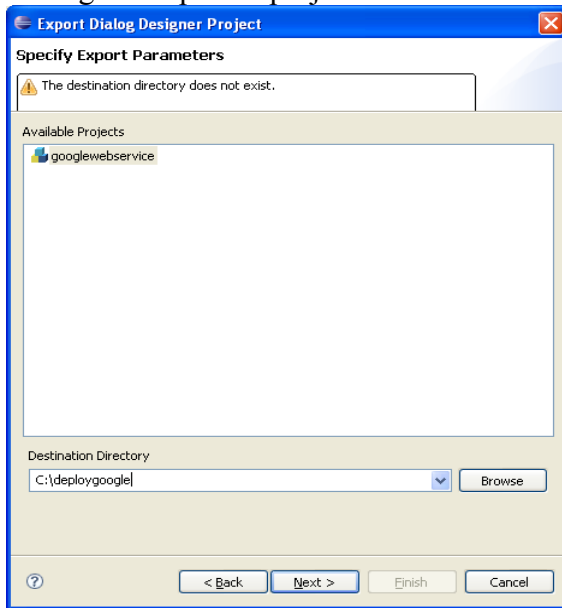
Step 4. Open the *File* menu and click on *Export* item as shown below.



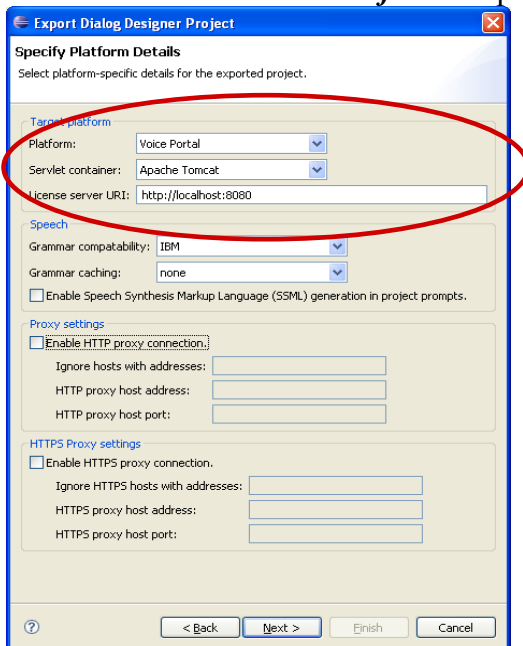
Step 5. Select the *Export Dialog Designer Project* under the *Avaya Speech Development* folder as shown below. Click the Next button to continue.



Step 6. Select an existing directory (using the **Browse** button) or create a new directory for storing the exported project file.

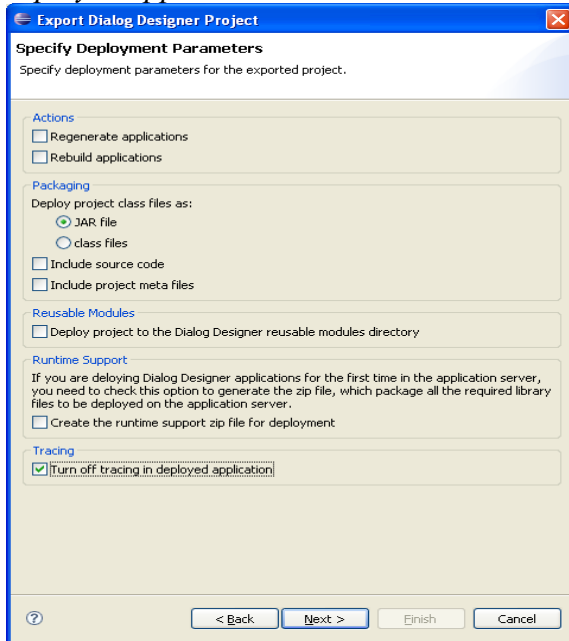


Step 7. Set the *Servlet Container*, *License URI* fields and the *Grammar Compatibility* fields as shown below. Depending on the desired platform, select either *Interactive Response* or *Voice Portal* from the **Platform** drop down box below. Click the Next button to continue.

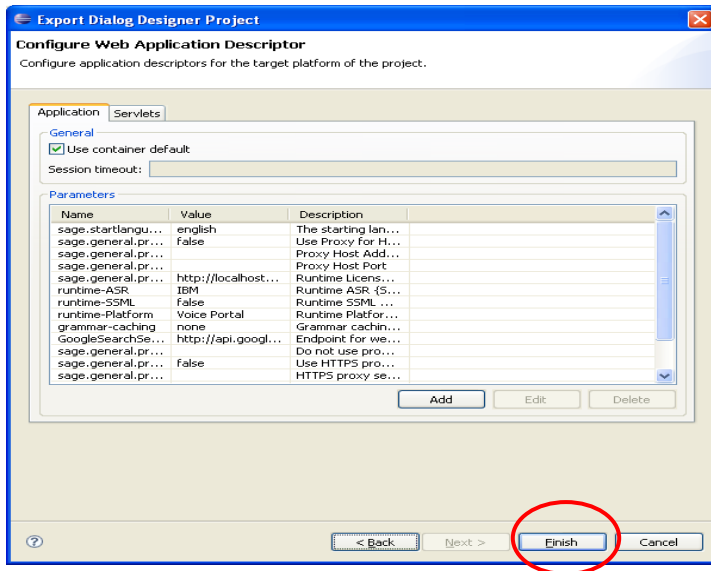


Step 8. Ensure that the settings on the form match the one's shown below.

Note: If tracing is needed for a deployed application, uncheck the *Turn off tracing in deployed application* check box.



Step 9. Review the settings as shown below and click the **Finish** button to generate the project file for deployment. Retrieve the file from the directory location specified in **Step 6** above.



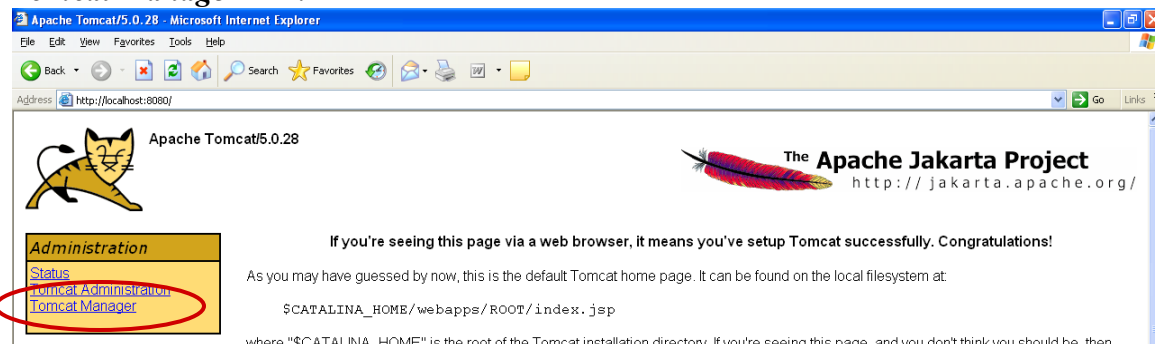
5. Deploying the *TelephonyWebService* Dialog Designer application on a local Tomcat Web Server

Note:

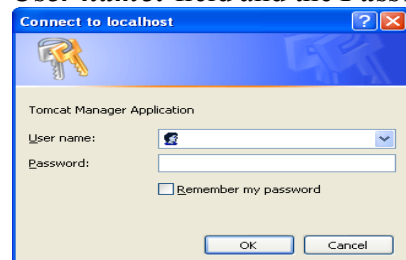
1. Delete the application and its Tomcat context definition from the Dialog Designer workspace before attempting to deploy the same application on the local Tomcat server using the below steps. Failure to do so will result in duplicate copies of the application on the same Tomcat server, resulting in unexpected behavior at runtime.
2. The Tomcat version may change depending on the Dialog Designer version installed. The procedure described below will work with Tomcat versions 5.0.28 and 5.5.23.
3. Locate the **tomcat-user.xml** file under *<Tomcat Installation directory>\conf* and add a manger role and a user for Tomcat manger as shown below:

```
<role rolename="manager"/>
<user username="<username>" password="<password>" roles="manager"/>
```
4. For a remote Tomcat (or other) web server the Dialog Designer Runtime files need to be installed prior to deploying the application.
5. These steps can be used to deploy any Dialog Designer application on a Tomcat web server.

Step 1. Access the local Tomcat web page by typing <http://localhost:8080>. Click on the **Tomcat Manager** link.



Step 2. Type in the user name and password created above in the **tomcat-users.xml** file in the **User name:** field and the **Password:** field. Click the **OK** button to login to the Tomcat manager.



Step 3. Scroll to the bottom of the page and locate the **WAR file to deploy** section. Click on the **Browse** button and select the **TelephonyWebService** exported project file generated by Dialog Designer. Click on the **Deploy** button to upload and deploy the application on Tomcat.

Path	Display Name	Running	Sessions	Commands
/	Welcome to Tomcat	true	0	Start Stop Reload Undeploy
/CPMClient-2.1.45	CPMClient	true	0	Start Stop Reload Undeploy
/CtiApp		true	0	Start Stop Reload Undeploy
/PPMAPI	Personal Profile Manager - Contact Lists	true	0	Start Stop Reload Undeploy
/TelephonyWebService		true	0	Start Stop Reload Undeploy
/WebLM	Avaya WebLM License Manager	true	0	Start Stop Reload Undeploy
/axis	Apache-Axis	true	0	Start Stop Reload Undeploy
/cticonnector		true	0	Start Stop Reload Undeploy
/host-manager	Tomcat Manager Application	true	0	Start Stop Reload Undeploy
/icconnector		true	0	Start Stop Reload Undeploy

Tomcat Version	JVM Version	JVM Vendor	OS Name	OS Version	OS Architecture
Apache Tomcat/5.5.23	1.5.0_09-b01	Sun Microsystems Inc.	Windows XP	5.1	x86

Copyright © 1999-2005, Apache Software Foundation

Step 4. Ensure that an **OK** message is displayed as shown below.

Note: A **OK- Deployed application from Context file: <pathname to war file>** message might also displayed

Path	Display Name	Running	Sessions	Commands
/	Welcome to Tomcat	true	0	Start Stop Reload Undeploy
/CPMClient-2.1.45	CPMClient	true	0	Start Stop Reload Undeploy
/CtiApp		true	0	Start Stop Reload Undeploy
/PPMAPI	Personal Profile Manager - Contact Lists	true	0	Start Stop Reload Undeploy
/TelephonyWebService		true	0	Start Stop Reload Undeploy
/WebLM	Avaya WebLM License Manager	true	0	Start Stop Reload Undeploy
/axis	Apache-Axis	true	0	Start Stop Reload Undeploy
/cticonnector		true	0	Start Stop Reload Undeploy
/host-manager	Tomcat Manager Application	true	0	Start Stop Reload Undeploy
/icconnector		true	0	Start Stop Reload Undeploy

Step 5. Click on the deployed application under the *Path* column as shown below.

The screenshot shows the Tomcat Web Application Manager interface. The browser window is titled '/manager - Microsoft Internet Explorer'. The address bar shows 'http://localhost:8080/manager/html/upload'. The page has a header with the Apache Software Foundation logo and a cat icon. Below the header, the title 'Tomcat Web Application Manager' is centered. A message box shows 'Message: OK'. Below the message box, there is a navigation bar with links: 'List Applications', 'HTML Manager Help', 'Manager Help', and 'Server Status'. The main section is titled 'Applications' and contains a table with columns: Path, Display Name, Running, Sessions, and Commands. The table lists several applications, including 'Welcome to Tomcat', 'CPMClient', 'CTIapp', 'Personal Profile Manager - Contact Lists', 'Avaya WebLM License Manager', 'Apache Axis', 'cticonnector', 'Tomcat Manager Application', and 'icconnector'. The row for 'Personal Profile Manager - Contact Lists' is circled in red.

Path	Display Name	Running	Sessions	Commands
/	Welcome to Tomcat	true	0	Start Stop Reload Undeploy
/CPMClient-2.1.45	CPMClient	true	0	Start Stop Reload Undeploy
/CTIapp		true	0	Start Stop Reload Undeploy
/PPMApp	Personal Profile Manager - Contact Lists	true	0	Start Stop Reload Undeploy
/TelephonyWebService		true	0	Start Stop Reload Undeploy
/WebLM	Avaya WebLM License Manager	true	0	Start Stop Reload Undeploy
/axis	Apache Axis	true	0	Start Stop Reload Undeploy
/cticonnector		true	0	Start Stop Reload Undeploy
/host-manager	Tomcat Manager Application	true	0	Start Stop Reload Undeploy
/icconnector		true	0	Start Stop Reload Undeploy

Step 6. If the application is deployed correctly then the page shown below will be displayed. Click on the *View* link to execute the Dialog Designer application in HTML mode.

The screenshot shows the Dialog Designer Application - TelephonyWebService page. The browser window is titled 'Dialog Designer Application - TelephonyWebService - Microsoft Internet Explorer'. The address bar shows 'http://localhost:8080/TelephonyWebService/'. The page has a title 'TelephonyWebService'. Below the title, there is a welcome message 'Welcome to Dialog Designer Application - TelephonyWebService.' and a section titled 'What do you want to do today?' with a bulleted list: 'Validate the application dependencies.', 'View the application in HTML mode. (For debugging purposes)', and 'Performance Monitoring for the application.' The 'View' link in the second bullet is circled in red. Below the list, there is a note: 'If you have problems getting the Dialog Designer application to work, contact Avaya through your normal support channels.'

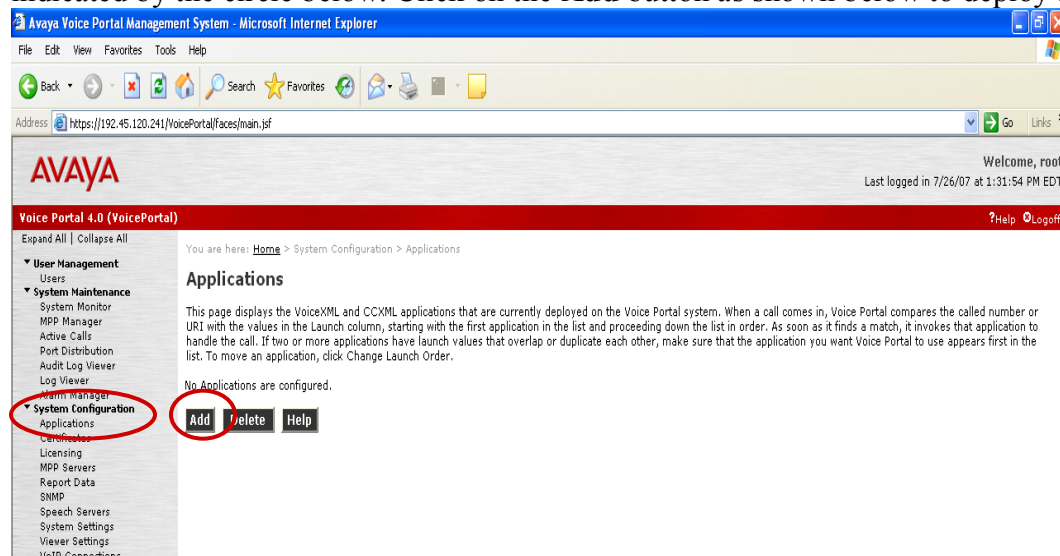
6. Deploying the *TelephonyWebService* application

6.1. Avaya Voice Portal Platform

Step 1. Access the Voice Portal Management System (VPMS) login page by typing *http://<IP Address of VPMS> /VoicePortal*. Accept the security certificate (not shown). Enter a user name in the **User Name** field and password in the **Password** field. Click on the **Submit** button as shown below.



Step 2. Click on the **Applications** link (under **System Configuration**) in the left hand pane as indicated by the circle below. Click on the **Add** button as shown below to deploy an application.



Step 3. Enter an appropriate name in the **Name** field. Select **VoiceXML** from the **MIME Type** drop down box. Enter the URL to the application in the **VoiceXML URL** field as shown below. The format is *http://<IP address of Tomcat Server hosting the application>:8080/TelephonyWebServices (Case sensitive)/Start*. Ensure that the Tomcat server hosting the application is started.

Click on the **Verify** button and view the application start up page displayed below. This step indicates that the application is deployed correctly.

Under the **Speech Servers** section, select a speech engine from the **ASR** drop down box and a TTS engine from the **TTS** drop down box.

Under the **Application Launch** section, check the **Inbound** radio button for the **Type** field as shown below. Check the **Number** radio button and type in the Voice Portal Hunt group pilot number in the **Called Number** field (not shown). Click on the **Add** button to add the Hunt group pilot number as shown below.

The screenshot displays the Avaya Voice Portal Management System interface in a Microsoft Internet Explorer browser. The main window shows the 'Change Application' configuration page. The 'Name' field is set to 'ICtestapp', the 'MIME Type' is 'VoiceXML', and the 'VoiceXML URL' is 'http://192.45.100.238:8080/TelephonyWebService/Start?ddtrace=true'. The 'Verify' button is circled in red. The 'Speech Servers' section shows 'ASR' set to 'Nuance' and 'TTS' set to 'Nuance'. The 'Application Launch' section shows 'Type' set to 'Inbound' and 'Number' selected. The 'Called Number' field contains '79403'. A secondary browser window shows the 'Form Start' page with the Avaya logo and application startup parameters.

Note: Refer to [3] for details on configuring Voice Portal extensions on the Avaya Communication Manager.

Step 4. Scroll to the bottom of the page and click on the **Save** button to save the application settings.

Avaya Voice Portal Management System - Microsoft Internet Explorer

Address: https://192.45.120.241/VoicePortal/faces/main.jsf

Welcome, root
Last logged in 10/17/07 at 5:33:16 PM EDT

Voice Portal 4.0 (VoicePortal)

Expand All | Collapse All

- ▼ **User Management**
 - Users
- ▼ **System Maintenance**
 - System Monitor
 - MPP Manager
 - Active Calls
 - Port Distribution
 - Audit Log Viewer
 - Log Viewer
 - Alarm Manager
- ▼ **System Configuration**
 - Applications
 - Certificates
 - Licensing
 - MPP Servers
 - Report Data
 - SNMP
 - Speech Servers
 - System Settings
 - Viewer Settings
 - VoIP Connections
- ▼ **Reports**
 - Application Summary
 - Application Detail
 - Call Summary
 - Call Detail
 - Performance
 - Session Summary
 - Session Detail

MIME Type: VoiceXML

VoiceXML URL: http://192.45.100.238:8080/TelephonyWebService/Start?ddtrace=true **Verify**

Speech Servers

ASR: Nuance TTS: Nuance

Languages: English(USA) en-us Voices: English(US) en-US Jennifer F

Application Launch

Type: ☒ Inbound ☐ Inbound Default ☐ Outbound

☒ Number ☐ Number Range ☐ URI

Called Number: 79403 **Add**

Remove

Speech Parameters

Reporting Parameters

Advanced Parameters

Save **Apply** **Cancel** **Help**

Step 5. The **Applications** page will be displayed indicating that the application is now successfully deployed.

Avaya Voice Portal Management System - Microsoft Internet Explorer

Address: https://192.45.120.241/VoicePortal/faces/main.jsf

Welcome, root
Last logged in 10/17/07 at 5:33:16 PM EDT

Voice Portal 4.0 (VoicePortal)

Expand All | Collapse All

- ▼ **User Management**
 - Users
- ▼ **System Maintenance**
 - System Monitor
 - MPP Manager
 - Active Calls
 - Port Distribution
 - Audit Log Viewer
 - Log Viewer
 - Alarm Manager
- ▼ **System Configuration**
 - Applications
 - Certificates
 - Licensing
 - MPP Servers
 - Report Data
 - SNMP
 - Speech Servers
 - System Settings
 - Viewer Settings
 - VoIP Connections
- ▼ **Reports**
 - Application Summary
 - Application Detail
 - Call Summary
 - Call Detail
 - Performance
 - Session Summary
 - Session Detail

You are here: [Home](#) > [System Configuration](#) > [Applications](#)

Applications

This page displays the VoiceXML and CCXML applications that are currently deployed on the Voice Portal system. When a call comes in, Voice Portal compares the called number or URI with the values in the Launch column, starting with the first application in the list and proceeding down the list in order. As soon as it finds a match, it invokes that application to handle the call. If two or more applications have launch values that overlap or duplicate each other, make sure that the application you want Voice Portal to use appears first in the list. To move an application, click Change Launch Order.

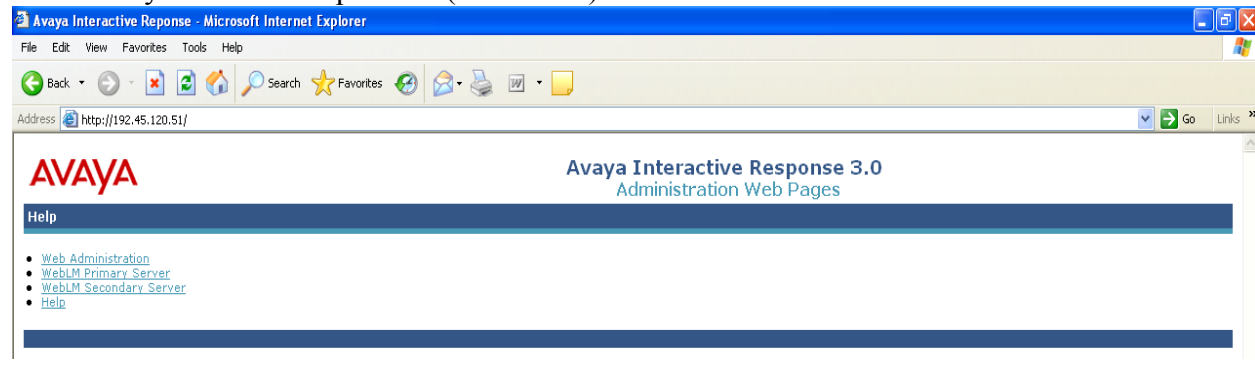
The information that you entered has been saved.

	Name	MIME Type	URL	Launch	ASR	Languages	TTS	Voices
☐	TelephonyWebServices	VoiceXML	http://192.45.100.238:8080/TelephonyWebService/Start?ddtrace=true	79403	Nuance	English(USA) en-us	Nuance	English(US) en-US Jennifer F

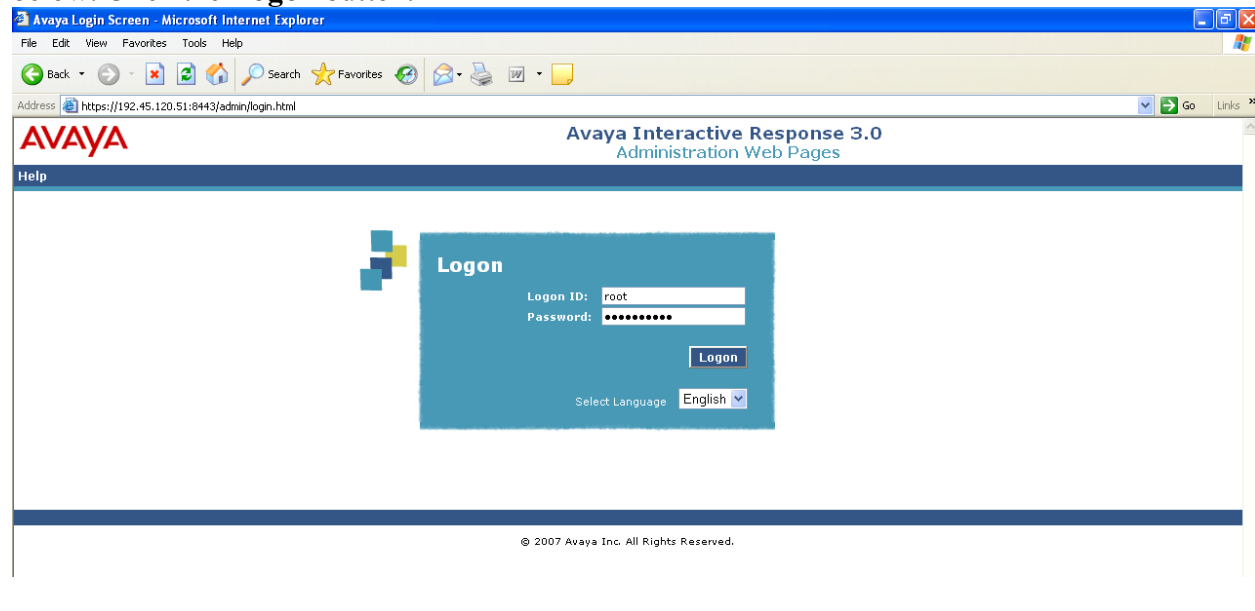
Add **Delete** **Help**

6.2. Avaya Interactive Response

Step 1. Access the Avaya Interactive Response web page by typing *http://<IP Address of Avaya Interactive Response>*. Click on the **Web Administration** link as shown below. Accept the security certificate to proceed (not shown).



Step 2. Enter a user name in the *Username* field and a password in the *Password* field as shown below. Click the **Logon** button.



Step 3. Once logged in, click on the *Channel Services* link under *Voice Services*. Select *Channel 0* under the *Channel Services* table by clicking on the checkbox beside the channels under the *Select* column. Click the **Assign Selected** button to assign an application to this channel.

Avaya IR Administration - Microsoft Internet Explorer

Address: https://192.45.120.51:8443/admin/admin.html

Avaya Interactive Response 3.0
Administration Web Pages

You are Here: Configuration Management>Voice Services>Channel Services

Channel Services

Select	Chan	Service/URI	Type	Startup Service/URI	Type
☒	0	-	unassigned	-	unassigned
☐	1	-	unassigned	-	unassigned
☐	2	-	unassigned	-	unassigned
☐	3	-	unassigned	-	unassigned
☐	4	-	unassigned	-	unassigned
☐	5	-	unassigned	-	unassigned
☐	6	-	unassigned	-	unassigned
☐	7	-	unassigned	-	unassigned
☐	8	-	unassigned	-	unassigned
☐	9	-	unassigned	-	unassigned

< Prev Channel Range: (0-9) Next > Display 10 channels.

Unselect All Assign Selected Unassign Selected Refresh

Step 4. Select *VXML URI* as the value for the *Assign* field as shown below. Click the **Submit** button to proceed.

Avaya IR Administration - Microsoft Internet Explorer

Address: https://192.45.120.51:8443/admin/admin.html

Avaya Interactive Response 3.0
Administration Web Pages

You are Here: Configuration Management>Voice Services>Channel Services>Assign Services to Channels

Assign Services to Channels

Assign: TAS Service

Service: TAS Service

Startup Service: VXML URI

To Chan(s): 54

Submit Reset Cancel Help

Step 5. Enter the complete path to the application under the **URI** field shown below. The complete path is specified by `http://<IP address of local machine>:8080/TelephonyWebServices/Start`. Click the **Verify** button after adding the application link in the **VoiceXML URL** field. The HTML test page for the application should be seen as shown below. This ensures that the Dialog Designer application is hosted correctly and the Dialog Designer licenses can be accessed. Ensure that the selected channels are displayed in the **To Chan(s)** field. Click on the **Submit** button to apply the changes.

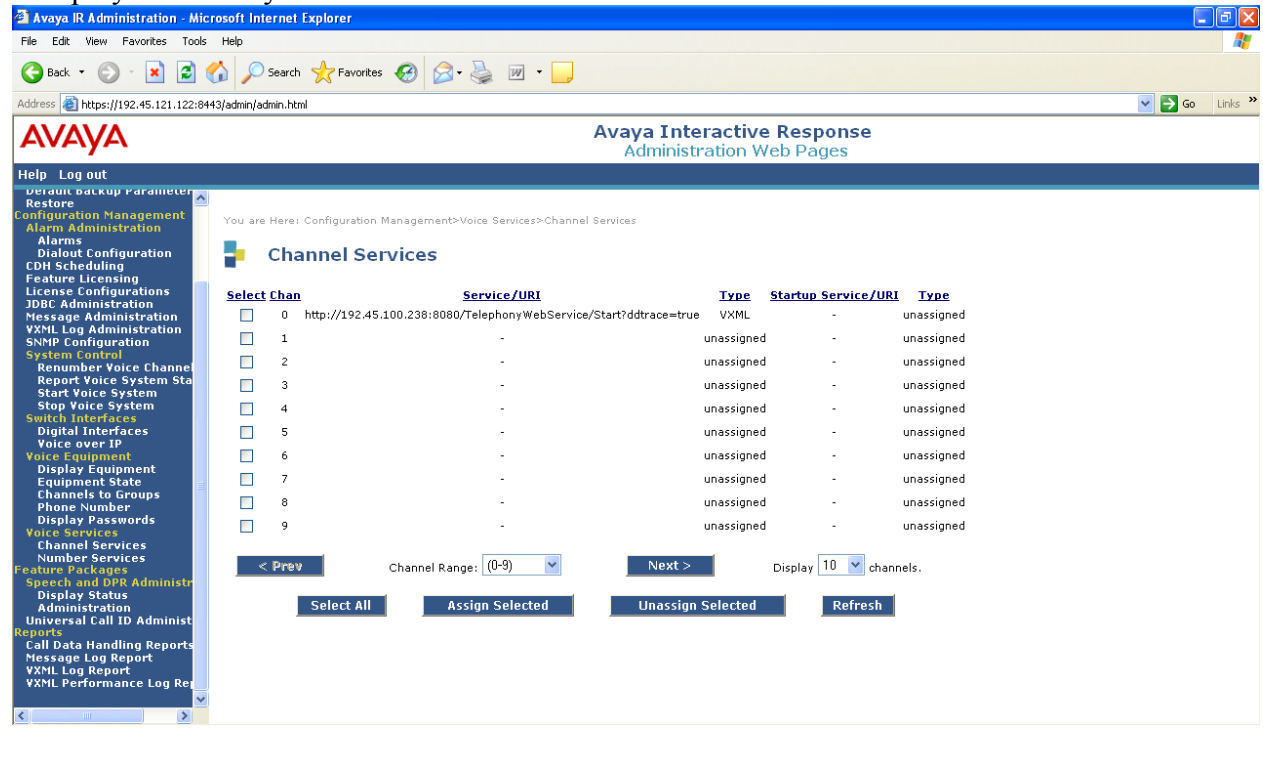
Note: Ensure that the Tomcat server hosting the application is started.

The screenshot shows the Avaya Interactive Response Administration Web Pages in a Microsoft Internet Explorer browser. The main window displays the 'Assign Services to Channels' form. The 'Assign:' dropdown is set to 'VXML URI'. The 'URI:' field contains '0.238:8080/TelephonyWebService/Start?ddtrace=true'. The 'Verify' button is circled in red. The 'To Chan(s):' field is empty. Below the form are buttons for 'Submit', 'Reset', 'Cancel', and 'Help'.

A secondary browser window is open, showing the XML test page at 'https://192.45.121.122:8443/servlet/TestURLConnection?UR...'. The XML content is as follows:

```
<?xml version="1.0" encoding="UTF-8" ?>
- <?xml version="2.0"
  xmlns="http://www.w3.org/2001/vxml" xml:lang="en-us"
  application="AppRoot?_DDSESSIONID=F2B9CEDEA25DBEB3EC66888B49ABF6FC3A%2FTelephonyWebService">
  <meta name="author" content="Avaya Dialog Designer" />
  <meta name="runtime-version" content="04.00.07" />
  <meta name="runtimecommon-version" content="04.00.07" />
  <meta name="copyright" content="Copyright (c) 2003-2007, Avaya" />
- <form id="Start">
  <script>function getRedirect( item ) { try { var len
    = session.connection.redirect.length; var result
    = ""; for (var i = 0; i < len; i++) { if (i > 0)
    { result += " |"; } var temp =
    "session.connection.redirect[" + i + "]."; item;
    result += eval(temp); } return result; } catch
    (errMsg) { return("unknown"); } } function
    getScriptItem( item ) { try { var result = "";
    result += eval(item); return result; } catch
```

Step 6. The **Channel Services** page is displayed as shown below indicating that the application is deployed correctly.



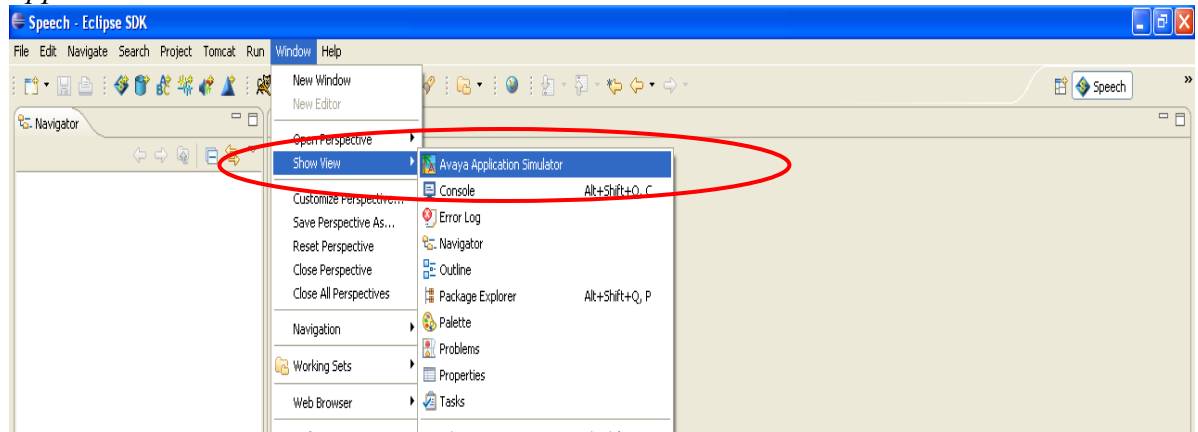
6.3. Testing the *TelephonyWebService* application

Note:

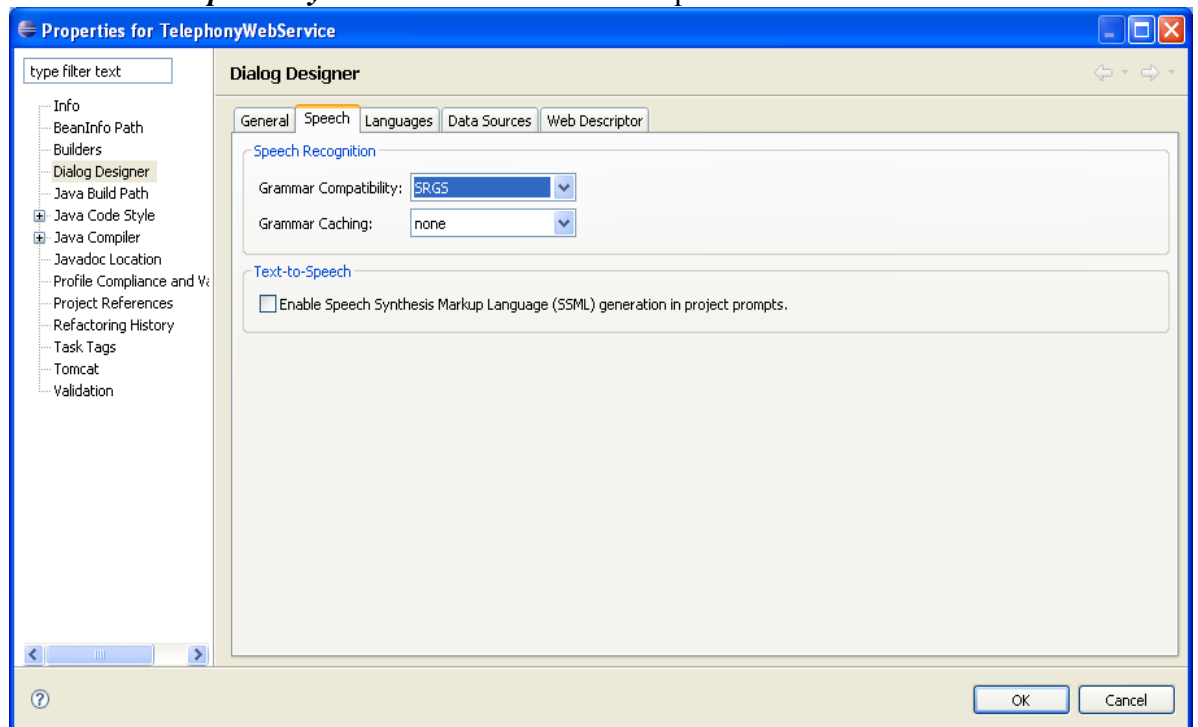
1. To test the application using a phone, place a call to the Voice Portal pilot number under the Launch column as shown in Step 5 of Section 6.1 above. In case of the IR, call the extension corresponding to the channel number on which the application is deployed.
2. The Avaya Application Simulator uses Microsoft Speech Recognizer and Microsoft SAPI Text-to-Speech to interact with the caller.
3. Certain call-flows involving CTI and IC components cannot be completely tested using the Avaya Application Simulator. Refer to [4] for additional details.

The steps below describe the process of testing the *TelephonyWebService* application using the built in Dialog Designer Avaya Application Simulator.

Step 1. The default Speech Perspective should display the Avaya Application Simulator as one of the tabs on the bottom section of the screen as shown below. If the simulator tab is not enabled by default then click on *Window* menu option and select *Show View* and *Avaya Application Simulator*.



Step 2. Right Click on the project in the *Navigator* view and select *Properties* in the drop down menu (not shown). Select Dialog Designer *Speech* Tab as shown below. Set the **Grammar Compatibility** field to *SRGS* from the drop down box.



Step 3. Click on the *Web Descriptor* tab and select the **runtime-parameter** field (Not shown). Click the **Edit** button (Not shown) and set the **value** field to *Desktop*. Click the **OK** button on all the open forms to save the changes.

Edit context parameter

Enter the values for the application context parameter.

Name: runtime-Platform

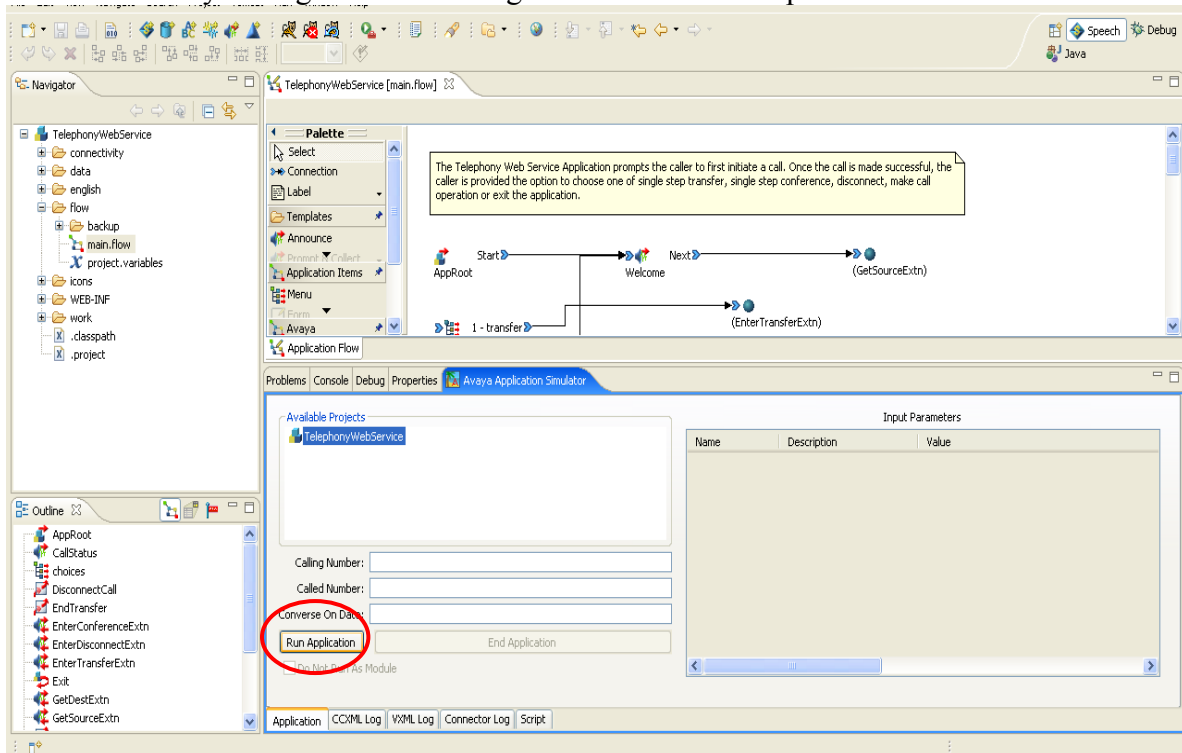
Value: Desktop

Description: Runtime Platform {Desktop|Voice Portal|Other|IR}

OK Cancel

Step 4. The Avaya Application Simulator automatically displays the projects imported into the Dialog Designer (Eclipse) environment. The screen below shows the **TelephonyWebServices** project listed in the *Application* tab of the simulator. Click on the **Run Application** button to start the call flow for this application. Refer to [4] on additional simulator inputs to this application

Note: The Proxy settings must be configured as shown in Step 3 of Section 4 above.



Step 5. The call flow executes as shown in the below window. The inputs to the application can be provided by using a microphone or through the key pad (depending on grammars supported by the application).

The screenshot shows the Eclipse IDE interface for the 'Speech - TelephonyWebService [main.flow]' project. The main workspace displays the 'Application Flow' diagram, which includes a 'Start' node, a 'Welcome' node, and a 'Next' node leading to '(GetSourceExtn)'. A '1 - transfer' node also leads to '(EnterTransferExtn)'. The 'Avaya Application Simulator' window is highlighted with a red circle. It features a numeric keypad, a 'Call Active' status indicator, and a log of events. The log shows the following sequence of events:

- Call Active
- Waiting ASR
- Waiting DTMF
- Playing text: "Welcome to Avaya telephony webservice sample application. Enter the source and destination extensions to initiate a call.", Language (en-us), Gender (Female)
- Playing text: "Please enter the source extension.", Language (en-us), Gender (Female)
- Waiting for DTMF input

Below the log, there are buttons for 'Send ASR', 'Record End', and 'Hang Up'. The 'ASR Result' and 'Call Result' dropdowns are also visible. The bottom of the simulator window shows a tabbed interface with 'Application', 'CCXML Log', 'VXML Log', 'Connector Log', 'Script', and 'Call::uid_12'.

7. References

The following documents can be downloaded from <http://support.avaya.com>

- [1] Application Enablement Services, Web Services Programmers Guide, Release 4.0
02-300362, Issue 3.0, March 2007
- [2] Administrators Guide to Avaya Communication Manger
03-300509, Issue 2.1, May 2006
- [3] Configuration Note 3910 – Rev E (10/07)
Avaya Voice Portal, Software Application
- [4] Avaya Dialog Designer
Dialog Designer Developers Guide
August 2007, Issue 1

©2012 Avaya Inc. All Rights Reserved.

Avaya and the Avaya Logo are trademarks of Avaya Inc. All trademarks identified by ® and ™ are registered trademarks or trademarks, respectively, of Avaya Inc. All other trademarks are the property of their respective owners. The information provided in these Application Notes is subject to change without notice. The configurations, technical data, and recommendations provided in these Application Notes are believed to be accurate and dependable, but are presented without express or implied warranty. Users are responsible for their application of any products specified in these Application Notes.

Please e-mail any questions or comments pertaining to these Application Notes along with the full title name and filename, located in the lower right corner, directly to the Avaya DevConnect Program at devconnect@avaya.com.